



Large Language Models in Data Science

Week 6: Hackathon – Applied LLM Projects

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Session Overview

Morning (Hackathon briefing)

1. Recap of course concepts (Weeks 1–5)
2. Hackathon rules and grading
3. Project scope: what fits into one day
4. Team formation and logistics

Full-day Hackathon

- ▶ Choose a problem and dataset
- ▶ Design an end-to-end LLM / RAG pipeline
- ▶ Implement, test, and iterate
- ▶ Prepare a short live demo and summary

Why a Hackathon?

- ▶ Move from **toy examples** to an **end-to-end project** in a realistic setting.
- ▶ Combine techniques from the whole course: tokenization, embeddings, HF models, prompting, classification, RAG.
- ▶ Practice **teamwork** around LLMs and data: dividing roles, coordinating experiments, merging ideas.
- ▶ Experience trade-offs between ambition and scope under strict time constraints.

Evaluation & Final Grade

- ▶ The hackathon is the **final evaluation** of the course.
- ▶ Final score out of 20 is based on four criteria (5 points each), as in the README:
 - ▶ **Use Case & Applications (5 pts)**: relevance, originality, potential future usefulness.
 - ▶ **Code Cleanliness (5 pts)**: readability, organization, modularity, reproducibility.
 - ▶ **Lecture Integration (5 pts)**: use of course concepts (prompting, RAG, embeddings, classification, agent loops, etc.).
 - ▶ **Presentation (5 pts)**: clarity, conciseness, visuals, explanation of design choices.
- ▶ Your hackathon score is your **official course grade**.

Teams & Logistics

- ▶ Work in groups of **1 to 5** students (recommended: 3–4).
- ▶ Each team produces:
 - ▶ a small but coherent **project repository or notebook(s)**,
 - ▶ a baseline comparison (quality, time, and API cost),
 - ▶ a short **live demo** of the system (or a recorded run, if needed),
 - ▶ a **5–7 minute presentation** covering problem, approach, and lessons learned.
- ▶ Use any combination of tools seen in the course:
 - ▶ Hugging Face, Gemini API, scikit-learn, sentence-transformers, simple RAG stacks.
- ▶ You are encouraged to reuse and adapt code from the labs (with proper structuring and documentation).

What Makes a Good Project?

- ▶ **Clear problem statement:** for whom is the system built, and what decision does it help?
- ▶ **Grounded in data:** use real or realistic data relevant to AMU, Marseille, or data science practice.
- ▶ **Course concepts in action:**
 - ▶ thoughtful prompts, embeddings, RAG, or classification models,
 - ▶ not just a UI wrapped around an API.
- ▶ **End-to-end story:** ingestion → processing / retrieval → model → output.
- ▶ **Simple but solid:** small scope, but robust, well-structured, and easy to demo.

Scope for One Day

- ▶ Aim for a **minimal viable assistant**, not a full product.
- ▶ Prefer:
 - ▶ one or two well-implemented workflows,
 - ▶ over many incomplete features.
- ▶ Reuse:
 - ▶ lab pipelines (classification, RAG) as starting points,
 - ▶ prompts and templates tested earlier in the course.
- ▶ Keep infrastructure light:
 - ▶ local notebooks, small datasets, API calls,
 - ▶ no need for full web deployment if time is short.

Idea 1: AMU Data Science Programme Assistant

- ▶ **Goal:** help prospective or current students explore the MAS / Data Science track at AMU.
- ▶ **Data:**
 - ▶ official programme pages (like those used in the Week 5 lab),
 - ▶ additional documents: FAQs, course descriptions, internship info.
- ▶ **Pipeline:**
 - ▶ ingest and chunk web pages / PDFs,
 - ▶ build a BM25 + embedding index with metadata (campus, year, language),
 - ▶ use a RAG chain to answer questions about courses, prerequisites, careers.
- ▶ **Extensions:**
 - ▶ multilingual Q&A (French / English),
 - ▶ citations and links back to source sections,
 - ▶ a small evaluation set of typical student questions.

Idea 2: Data Science EDA Copilot

- ▶ **Goal:** assist a data scientist in exploring a tabular dataset (e.g., churn, housing, or a public Kaggle dataset).
- ▶ **Workflow:**
 - ▶ user provides a CSV and a question (e.g., “What drives churn in this dataset?”),
 - ▶ the system generates Python code snippets for plots / tables using prompting techniques from Week 3,
 - ▶ execute the code, collect results, and ask the LLM to summarise findings in natural language.
- ▶ **Course concepts:**
 - ▶ prompt templates for code generation and summarisation,
 - ▶ structured outputs (e.g., JSON with “steps”, “code”, “insights”),
 - ▶ maybe a simple classification model from Week 4 as a baseline.
- ▶ **Focus:**
 - ▶ reproducibility (scripts/notebooks can be re-run),
 - ▶ clear separation between data, code, and LLM outputs.

Idea 3: Student Support Inbox Triage

- ▶ **Goal:** help route incoming student emails to the right service and provide draft replies.
- ▶ **Data:**
 - ▶ a synthetic dataset of student enquiries (motivated by Week 4 intent classification),
 - ▶ labels such as `timetable`, `exam`, `administrative`, `internship`, `other`.
- ▶ **Pipeline:**
 - ▶ baseline classifier (embeddings + logistic regression or zero-shot classification),
 - ▶ optional RAG component with a small FAQ corpus,
 - ▶ LLM prompting to generate a draft response and a recommended action.
- ▶ **Evaluation:**
 - ▶ accuracy of routing to labels (using a held-out test set),
 - ▶ qualitative assessment of suggested replies (are they precise and polite?).

Checklist for Today

- ▶ **1. Define the problem:**
 - ▶ target user, key questions, success criteria.
- ▶ **2. Select data & models:**
 - ▶ datasets, external documents, HF models, LLM endpoints.
- ▶ **3. Design the pipeline:**
 - ▶ retrieval / preprocessing, model calls, post-processing.
- ▶ **4. Implement the minimal version:**
 - ▶ one or two key workflows working end-to-end.
- ▶ **5. Polish & present:**
 - ▶ clean up code, add README, prepare a concise demo.

Final Remarks

- ▶ Use this day to **experiment** and **connect ideas** from the course.
- ▶ Keep the scope realistic, but do not hesitate to be creative in your design.
- ▶ Remember the evaluation criteria: clarity of use case, clean code, integration of course content, and a clear presentation.
- ▶ Most importantly: have fun building with LLMs and data.